Haozhe Liu

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Singapore

Research Scientist and **founding member of NVIDIA Research Singapore Lab**. As **Project Lead of SoL-Pi** and a **Core Contributor to the SANA and SoL series**, I study recursive self-improvement for agent harnesses and inference acceleration. Before NVIDIA, I worked at Meta on multimodal foundation models and generative AI products, and earned my Ph.D. at KAUST with **Jürgen Schmidhuber**. My experience spans **1K+ GPU pretraining**, billion-pair datasets, and **20B-parameter models**. I have co-authored **40+ research papers** with **1.8K+ citations**, received a **Best Paper Award at the NeurIPS 2023 R0-FoMo Workshop**, and led or contributed to open-source projects with a combined **10K+ GitHub stars**. In addition, the open-source models I led have reached **600K+ downloads on Hugging Face**.

Research Experience

NVIDIA Research, Efficient AI — Singapore
Research Scientist, working with Prof. Song Han — *2026 – present*
Founding member of NVIDIA Research Singapore Lab.

- **Project Lead of SoL-Pi.** Scale auto-research loops to discover reusable harness improvements for more token-efficient coding agents.
- **Project Lead of SoL-Refiner.** Build one-step 4K video refinement with a **27× speedup** on MiniMax H3; its open release, **Open-H3-Regenerate-2K**, has **600K+ community downloads** on Hugging Face.
- **Core Contributor to SoL-Series.** Advance **RSI for Inference Acceleration** through **SoL-Engine** and **SoL-Attn**, developing agentic acceleration workflows for video generation models.
- **Core Contributor to SANA-Series.** Contribute to **SANA-WM**, **SANA-Video-2.0**, and **SANA-Streaming** across world modeling, efficient video generation, and real-time streaming.

Meta, BizAI — MPK, United States
Research Scientist Intern, with Ding Liu and Wei Liu — *2025*

- Co-developed a commercial **7B text-to-image model**, trained on over **1B image–text pairs** and deployed in production.
- Built a **20B native LLM-driven multimodal generation prototype** from scratch.
- Contributed to unified visual understanding and generation models and the **Neural Computers** project.

Meta, MGenAI — London, United Kingdom
Research Scientist Intern, with Juan-Manuel Pérez-Rúa and Tao Xiang — *2024*

- Co-developed a text-to-image foundation model supporting widely used products.
- Scaled autoregressive diffusion for video generation through **MarDini**.

KAUST, AI Initiative — Saudi Arabia
Ph.D. Researcher, advised by Prof. Jürgen Schmidhuber — *2022 – 2026*

- Studied multi-step inference in diffusion models, autoregressive models, and RL agents.
- Developed **T-GATE** attention caching, merged into Hugging Face Diffusers, and co-developed **NLSOM** for collective intelligence among agents.

Tencent, Jarvis Lab — Shenzhen, China
Research Intern, with N. He, Y. Li, and Y. Zheng — *2021 – 2022*

- Conducted research on generative models and medical imaging.

Research Mentorship

Tian Ye (Ph.D. candidate at HKUST; co-mentored with Dr. Enze Xie and Duomin Wang)
Research topics: Recursive self-improvement, efficient agent harnesses (SoL-Pi), and efficient video generation.

Haoyi Zhu (Ph.D. student at USTC; co-mentored with Dr. Enze Xie)
Research topics: World models and controllable video generation (SANA-WM).

Qihang Cao (Ph.D. student at NTU)
Research topics: Automated research and coding-agent harness efficiency (SoL-Pi).

Sensen Gao (Ph.D. student at NTU)
Research topics: Automated research and coding-agent harness efficiency (SoL-Pi).

Yitong Li (Ph.D. student at University of Hong Kong; co-mentored with Dr. Enze Xie)
Research topics: Model inference acceleration and agent harnesses (SoL-Engine, SoL-Pi).

Lei Zhang (Ph.D. student at UC San Diego; co-mentored with Dr. Hanrong Ye)

Selected Publications

Selected work by research direction. Full list: Google Scholar.

Scalable Agentic Systems

1. **SoL-Pi: Recursively Scaling Auto-Research Loops for Efficient Agent Harness**
   Haozhe Liu, T. Ye, S. Gao, et al. · **Technical report, 2026** [Paper] [Project] [Code]
2. **Learning to Identify Critical States for Reinforcement Learning from Videos**
   Haozhe Liu, M. Zhuge, B. Li, et al. · **ICCV 2023** [Paper] [Code]
3. **Mindstorms in Natural Language-Based Societies of Mind**
   M. Zhuge, **Haozhe Liu**, F. Faccio, et al. · **Computational Visual Media, 2025** · **Equal contribution** [Paper] [Code]
   NeurIPS 2023 R0-FoMo Workshop (Oral, Best Paper) [Workshop Paper]
4. **Neural Computers**
   M. Zhuge, C. Zhao, **Haozhe Liu**, et al. · **Technical report, 2026** · **Main contributor** [Paper] [Code]

Foundation Model Development

5. **Mixture of States: Routing Token-Level Dynamics for Multimodal Generation**
   Haozhe Liu, D. Liu, M. Zhuge, et al. · **CVPR 2026** [Paper] [Project]
6. **MarDini: Masked Autoregressive Diffusion for Video Generation at Scale**
   Haozhe Liu, S. Liu, Z. Zhou, et al. · **TMLR 2025** [Paper] [Project]
7. **SANA-WM: Efficient Minute-Scale World Modeling with Hybrid Linear Diffusion Transformer**
   H. Zhu, **Haozhe Liu**, Y. Zhao, et al. · **Technical report, 2026** · **Equal contribution** [Paper] [Project] [Code]
8. **TUNA: Taming Unified Visual Representations for Native Unified Multimodal Models**
   Z. Liu, W. Ren, **Haozhe Liu**, et al. · **CVPR 2026** · **Main contributor** [Paper] [Project]
9. **SANA-Video 2.0: Hybrid Linear Attention with Attention Residuals for Efficient Video Generation**
   J. Chen, J. Yu, Y. Li, . . . , **Haozhe Liu**, et al. · **Technical report, 2026** [Paper] [Project] [Code]
10. **SANA-Video: Efficient Video Generation with Block Linear Diffusion Transformer**
    J. Chen, Y. Zhao, J. Yu, . . . , **Haozhe Liu**, et al. · **ICLR 2026 (Oral)** [Paper] [Project] [Code]

Selected Publications (continued)

Efficient AI

11. **SoL-Refiner: Speed-of-Light One-Step Refinement for High-Resolution Video**
Haozhe Liu, T. Ye, S. Xue, et al. · **Technical report, 2026** [Paper] [Project] [Code]

12. **Faster Diffusion Through Temporal Attention Decomposition**
Haozhe Liu, W. Zhang, J. Xie, et al. · **TMLR, 2025; presented at ICLR 2026** [Paper] [Code]

13. **Sol Video Inference Engine: Agent-Native Full-Stack Acceleration Framework for Efficient Video Generation**
Y. Li, J. Chen, H. Li, **Haozhe Liu**, et al. · **Technical report, 2026**
· **Equal contribution** [Paper] [Project] [Code]

14. **Sol-Attn: Accelerating Video Generation Inference via On-the-Fly Attention Sparsification**
H. Li, Y. Li, J. Chen, . . . , **Haozhe Liu**, et al. · **Technical report, 2026** [Paper] [Project] [Code]

15. **Adaptive Caching for Faster Video Generation with Diffusion Transformers**
K. Kahatapitiya, **Haozhe Liu**, S. He, et al. · **ICCV 2025** [Paper] [Project] [Code]

Earlier Research

16. **AdaptiveMix: Improving GAN Training via Feature Space Shrinkage**
Haozhe Liu, W. Zhang, B. Li, et al. · **CVPR 2023** [Paper] [Code]

17. **Combating Mode Collapse via Offline Manifold Entropy Estimation**
Haozhe Liu, B. Li, H. Wu, et al. · **AAAI 2023** [Paper] [Code]

18. **Group-wise Inhibition based Feature Regularization for Robust Classification**
Haozhe Liu, H. Wu, W. Xie, et al. · **ICCV 2021** [Paper] [Code]

19. **Dynamically Masked Discriminator for Generative Adversarial Networks**
W. Zhang, **Haozhe Liu**, B. Li, et al. · **NeurIPS 2023** · **Equal contribution** [Paper] [Code]

20. **Effective Presentation Attack Detection Driven by Face Related Task**
W. Zhang, **Haozhe Liu**, F. Liu, et al. · **ECCV 2022** · **Equal contribution** [Paper] [Code]

Education

King Abdullah University of Science and Technology (KAUST) Saudi Arabia
Ph.D. in Computer Science Aug. 2022 – Mar. 2026
Advisor: Prof. Jürgen Schmidhuber

Awards & Grants

Best Paper Award, NeurIPS R0-FoMo Workshop, for NLSOM 2023
Co-PI, "Efficient Generative Models", SDAIA–KAUST grant (200k SAR) 2023
4th place, ECCV NICO Challenge 2022